

Weekly Report: [W07]

General Information

- **Group ID:** 52
- **Group Name:** Group 52
- **Class:** 25A01
- **Project Name:** Super Mario Game
- **Date range:** 2026-07-19 – 2026-07-25
- **Github Repository:** <https://github.com/ndmhuy/SuperMarioGame.git>

Tasks Completed This Week

25125083 - Nguyễn Đình Minh Huy (Member A - Engine, Physics & Player/Item Entities)

Branch: `dev` (Integrated from `A/feature/save-load` & `A/feature/level-editor`)

- **Task 1: Feature Branches Integration & Code Review Fixes:** Merged task branches `A/feature/save-load` and `A/feature/level-editor` into `dev`. Addressed code review feedback for PR #5 & PR #6: fixed `EventBus` subscriber listener sentinel invalidation via `static_cast<size_t>(-1)`, replaced letter-based debug keys with `Num1` - `Num9`, and encapsulated keybindings accessors in `Game`.
- **Task 2: Achievement Toast Notification Queue Cleanup:** Resolved stale toast rendering bug on slot reloads inside `AchievementManager` by clearing the `m_activeToasts` queue in `setUnlockedAchievements()`, and aligned achievement string keys with `SPEC.md` schemas (`100_coins`, `beat_bowser`).
- **Task 3: Consolidated Unit Test Runner (`verify_all`):** Created a unified test harness `verify_all.cpp` combining Map Editor command undo/redo execution, level JSON serialization/deserialization, slot save/load persistence, statistics tracker event aggregation, settings save/load, and achievement unlocking conditions into a single 6-test suite. Updated `CMakeLists.txt` and verified 100% assertions pass.
- **Task 4: 2-Player & Shadow Mario Architecture Plan:** Authored detailed architectural design document `two_player_ai_plan.md` specifying Shared-Screen vs Split-Screen

viewports, AI Bot personalities (Speedrunner, Hunter, Collector), and Shadow Mario circular input buffer (`PlayerFramePacket` deque) for delayed ghost playback.

- **Task 5: Save/Load & Level Editor Integration Validation:** Ran interactive `SuperMarioGame` binary demo to verify real-time slot JSON generation (`slot_1.json` , `slot_2.json` , `slot_3.json`), settings persistence (`settings.json`), and seamless state switching between `PlayingState`, `MapEditor`, and `Pause HUD`.
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25125084 - Trần Gia Huy (Member B - Input, Sound & Gameplay Systems)

Branch: `dev` (Integrated from PR #8 `B/feature/entities`)

- **Task 1: Entity Factory & Concrete Entities Pipeline:** Implemented the Factory Pattern in `EntityFactory` to instantiate 25+ concrete entity types using JSON type tags and initial position vectors.
- **Task 2: Enemy Hierarchy & AI Movement Strategies:** Coded 11 concrete enemy classes (`Goomba`, `KoopaTroopa`, `KoopaParatroopa`, `Boo`, `PiranhaPlant`, `BulletBill`, `HammerBro`, `Thwomp`, `ChainChomp`, `Lakitu`, `Spiny`) wired to 8 distinct Strategy subclasses (`PatrolStrategy`, `FlyStrategy`, `ChaseStrategy`, `LinearStrategy`, `TimerEmergenceStrategy`, `HammerThrowStrategy`, `TetheredChaseStrategy`, `ProximityTriggerStrategy`).
- **Task 3: Interactive Blocks & Platforms:** Implemented 9 block and platform classes (`BrickBlock`, `QuestionBlock`, `Pipe`, `Flagpole`, `HiddenBlock`, `MovingPlatform`, `FallingPlatform`, `IceBlock`, `ConveyorBelt`).
- **Task 4: Gameplay Graphical HUD Overlay & Boss Health Bar:** Developed `Hud` rendering live player stats (lives, score, coin count, remaining time, world stage name) and dynamic boss health bar indicators. Integrated custom retro typography (`PressStart2P.ttf` , `SuperMario256.ttf`).
- **Task 5: SpriteSheet, Animator & Verification Targets:** Developed `SpriteSheet`, `AnimationManager`, and `Animator`. Created component unit test verification targets (`verify_blocks.cpp` , `verify_blocks_new.cpp` , `verify_enemies.cpp` , `verify_enemies_new.cpp` , `verify_graphics.cpp` , `verify_graphics_visual.cpp` , `verify_hud_boss.cpp`).
- **Task 6: Asset Tree Consolidation:** Unified duplicate `asset/` and `assets/` resource paths into a single structured root directory (`assets/`).

Branch: `B/feature/graphic-visual` (*Graphics & HUD Extensions*)

- **Task 1: Real-Time Minimap Overlay Component:** Coded `Minimap` component drawing scaled tilemap preview overviews, camera view frustums, and real-time player tracking dots.

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- **Task 2: Minimap Visual Test Target:** Created interactive test suite `verify_minimap_visual.cpp` verifying viewport transforms and real-time minimap rendering.
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AI Usage Declaration

- Used AI to resolve `EventBus` subscriber handle sentinel logic during array mutation when unsubscribing listeners during event dispatches.
 - Used AI to design a lightweight `verify_all.cpp` test harness combining unit test targets without external testing framework overhead.
 - Used AI to design the circular packet buffer architecture (`PlayerFramePacket` deque) for Shadow Mario input recording and playback in `docs/two_player_ai_plan.md` .
 - Used AI to structure parameter passing for 25+ concrete entities inside `EntityFactory` to satisfy non-default constructors.
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Tasks Planned for Next Week

- **Nguyễn Đình Minh Huy:** Begin Phase 7 (Game States, Main Menu screen, Options menu UI, and World Map node path selection graph). Implement Time Rewind circular buffer memento state snapshots.
 - **Trần Gia Huy:** Merge `B/feature/graphic-visual` to `dev` . Implement Phase 10 In-Game Debug Overlay panel & Command Console (`~` key toggle), and set up 2-Player split-screen viewport rendering in `PlayingState` .
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Issues & Resolutions

Issue 1: EventBus Subscriber Listener Sentinel Bug on Unsubscribe

- **Problem:** When unsubscribing an EventBus listener handle in `PlayingState` , invalid handle checks using raw integer comparisons caused out-of-bounds array access or missed listener removals.
- **Resolution:** Updated `EventBus` subscriber handle sentinel representation to `static_cast<size_t>(-1)` to safely handle invalid or unsubscribed listener IDs across event broadcasts.

Issue 2: Stale Toast Rendering on Achievement Reset / Reload

- **Problem:** Loading a saved slot or invoking `setUnlockedAchievements()` did not clear active visual toast notifications in `AchievementManager`, causing previously unlocked toasts to pop up continuously after reloading a save.
- **Resolution:** Added explicit `m_activeToasts.clear()` inside `AchievementManager::setUnlockedAchievements()` and `reset()` to flush stale toasts prior to loading new slot states.

Issue 3: Save Slot Out-of-Bounds Flagpole and Spawn Point Coordinates

- **Problem:** Saving game state when player coordinates were out-of-bounds or near edge boundaries wrote raw negative or out-of-range floats into save slot JSON files, causing crash or invalid position loading.
- **Resolution:** Added `MathUtils::clamp` coordinate verification inside `Serializer.cpp` and `LevelLoader.cpp` to bound spawn points and flagpole coordinates strictly within level dimensions before JSON serialization.

Issue 4: EntityFactory Constructor Mismatch for Concrete Entities

- **Problem:** Instantiating derived entity classes in `EntityFactory::createEntity` using default constructors (`make_unique<T>()`) failed to compile because `Mario`, `Luigi`, `Toad`, `Peach`, and item subclasses require explicit position parameter constructors (`sf::Vector2f position`).
- **Resolution:** Updated `EntityFactory` switch-case factory methods to pass the target `position` argument explicitly to every concrete class constructor.